Challenges in Modeling of BLDC Motors for Design of Lightweight Robotic Systems

Ung Hee Lee
Deaprtment of Mechanical Engineering, Robotics Institute
University of Michigan
Ann Arbor, MI
unghee@umich.edu

Elliott J. Rouse
Department of Mechanical Engineering, Robotics Institute
University of Michigan
Ann Arbor, MI
ejrouse@umich.edu

I. Introduction

Brushless Direct Current (BLDC) Motors (Fig. 1) are used over brushed direct current motors due to their improved efficiency, speed, reliability, longevity and reduced noise. Due to the absence of mechanical brushes, BLDC motors are compact and efficient actuation method which is attractive for robotic numerous systems.

To minimize actuator mass, motors must be properly matched for their application, which is especially important for lightweight robotic systems, such as wearable robots and legged robots. To appropriately size the motor, modeling analyses are complete *a priori* that investigates the motor torque-velocity and current-voltage requirements, as well as temperature rise of the windings and housing. To complete these analyses, parameters such as the torque constant, winding resistances are required, and are often obtained from manufacturer's datasheets.

Motor datasheets provides important information to select adequate motors for the application; however, the description or interpretation of it can lead to common errors in motor modeling. Typically, roboticists simplify the analysis by considering the BLDC motors as equivalent "brushed" motor due to its simplicity; however, there are common errors that are prone to this process. For example, it can be unclear how to convert terminal resistance of BLDC motors to the "brushed" motor model, because BLDC motors have three phases with different winding configurations. This ambiguity of conversion process can cause inaccurate estimation of thermal loss, which may lead to conservative or amplified assessment of the motor. Therefore, it is critical to provide the units or standardize the metrics provided in motor datasheets while accurately interpreting them, especially for applications that motor mass is a key design factor.

II. Implications in BLDC Analyses

A. BLDC datasheet variation

Inconsistent representations or ambiguous descriptions of motor parameters in datasheet can result in inaccurate analysis of BLDC motors. For example, torque constants are represented in different reference frame of currents: line, phase and q-axis currents. With improper combinations of the torque constants and current can lead to erroneous torque estimation.

B. Standardizing Motor Analysis

Motor manufacturers should provide BLDC motor parameters in consistent reference frames . If this is impractical, manufacturers should provide information such as winding configuration (*i.e.* wye or delta) and units of each motor parameters to convert to desired units.

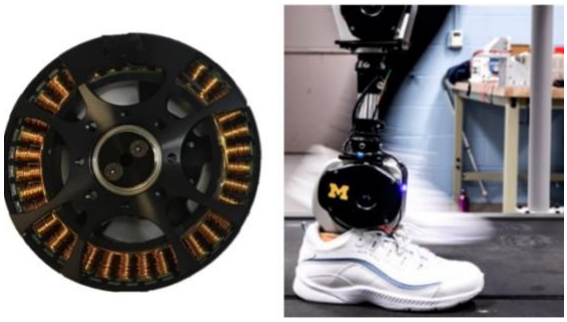


Figure 1. The picture of the high-torque exterior-rotor type Brushless DC motor (T-Motor 100 KV, left), and the University of Michigan's Open Source Leg which uses the T-motor for actuation (right).

We recommend using the q-axis reference frame for all motor analysis—this ensures proper modeling of the electro-mechanical requirements for a specific use case. Modeling in this way preserves the power loss calculations and ensures proper current-voltage and thermal responses. The following equations show torque and power of a BLDC motors represented in q-axis reference frame:

$$\tau = K_t^q I^q$$
$$P = I_q^2 R^q$$

where τ, K_t, I, R, P represents torque, torque constant, current, resistance and power, respectively, and q denotes q-axis quantities. These q-axis parameters are obtained from converting line-to-line or phase quantities by applying direct quadrature transformation, which changes the phase or line reference frame to that of the rotor frame. For example, the conversions of currents are:

$$\text{Wye: } \sqrt{1.5}\,\mathrm{I}^{\mathrm{line}} = \sqrt{1.5}\,\mathrm{I}^{\mathrm{phase}} = I^q$$
$$\text{Delta: } \sqrt{0.5}\,\mathrm{I}^{\mathrm{line}} = \sqrt{1.5}\,\mathrm{I}^{\mathrm{phase}} = I^q$$

where each conversion depends on the winding configuration of a motor. In this submission, we intend to provide a comprehensive overview of the modeling BLDC motors, which will follow an upcoming tutorial manuscript.

III. Conclusion

The intent of this work is twofold: i) to provide engineers who utilize BLDC motors with the tools to accurately model them and ii) highlight the need of standardization of representations in BLDC datasheets. We hope this work can be served as a benchmark for BLDC motor analyses and foster the use of BLDC motors in low-mass robotic applications.